

Quartic Causal Renormalization and Zero-Repair Inventory

Master Checkpoint – Invariant-Tube Edition

25 August 2026

Abstract

This document freezes the current state of the explicit quartic zero-repair program. The rigorous result already obtained is a logarithmic lower bound for the finite-horizon zero-repair bill, while the matching logarithmic upper bound remains open. Since the previous checkpoint, the obstruction has been localized much more sharply. The scalar quartic renormalization is explicit, the fresh phase sector created by block doubling is explicit, fixed-block causal phase feasibility has an exact polyhedral/Farkas description, and the reduced two-block profile problem admits a dimension-free local repair mechanism. Numerically, the fresh phase has a stable scale-covariant size and three bounded rank-one circulation modes give a well-conditioned local controller across the audited dyadic scales. The exact low-order fixed point $(7/9, 4/3, 6)$ is not dynamically invariant. The surviving mechanism is a *robust invariant tube*: a small neighborhood of the critical profile containing interior reachable centers with positive reduced Farkas margin.

The remaining theorem-level bridge is therefore not “find the fixed point” and not merely “bound a Hoffman constant.” It is a uniform, prefix-compatible *Safe-Anchor / Invariant-Tube Lemma* that preserves a phase-augmented critical tube under dyadic doubling and pays only a scale-independent amount of zero-repair inventory. Until that lemma is proved, no $O(\log n)$ upper bound and no $\Theta(\log n)$ theorem is claimed.

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1 Executive status

For the explicit quartic pair studied here,

$$\mathcal{B}_n = \Omega(\log n)$$

is rigorous. The desired matching upper bound

$$\mathcal{B}_n = O(\log n)$$

remains open. The current proof frontier is the implication

$$\text{uniform phase-safe dyadic continuation with bounded bill} \implies \mathcal{B}_{2n} \leq \mathcal{B}_n + C.$$

Iterating a bounded doubling increment would give

$$\mathcal{B}_{2^k} \leq \mathcal{B}_1 + kC = O(k),$$

and hence $\mathcal{B}_n = O(\log n)$ up to the usual dyadic interpolation. Combined with the proved lower bound, this would yield

$$\mathcal{B}_n = \Theta(\log n).$$

The current state can be summarized in one sentence:

the fixed point failed; the invariant-tube mechanism survived.

Statement	Status
Exact quartic source algebra and moment matching through order 3	PROVED
Exact two-step causal obstruction 5/8 versus static 7/8	PROVED
Exact reduction of the saturated two-step causal problem to an N^2 transport LP	PROVED
Exact quartic defect renormalization and critical condition $rR^2 = 2$	PROVED
Rigorous lower bound $\mathcal{B}_n = \Omega(\log n)$	PROVED
Exact kinematic phase decomposition and $\dim K_m = m^2$	PROVED
Exact fixed-block local causal phase polytope and Farkas criterion	PROVED
Dimension-free reduced-profile inball repair lemma	PROVED
Scale-covariant fresh-phase bounds on audited scales	NUMERICAL
Uniformly nonsingular bounded three-mode controller on audited scales	NUMERICAL
Positive reduced safe cubes / invariant-tube centers on audited scales	NUMERICAL
Uniform phase-safe anchor for every dyadic scale	OPEN
Uniform prefix-compatible inventory increment bound	OPEN
Matching $O(\log n)$ upper bound / $\Theta(\log n)$ law	OPEN
Exact asymptotic constant of $\mathcal{B}_n / \log n$	OPEN

2 The resource being priced: zero-repair inventory

The horizon parameter n in $\mathcal{B}_n$ is the depth of a causal history tree. It is *not* the numerical support length $N = 65, 129, 257, \dots$ used in the renormalization experiments.

Let

$$\mu_P = \sum_{i=1}^r p_i \delta_{x_i}, \quad p_i > 0, \quad \sum_i p_i = 1,$$

and let μ_Q be a finite-support probability law on $[0, \infty)$.

Definition 1 (Depth- n exact zero-repair tree). *A depth- n zero-repair tree is a family of buffer laws*

$$\{B_h\}_{|h| \leq n}$$

indexed by source histories such that every internal node satisfies

$$\boxed{\mu_Q * B_h = \sum_{i=1}^r p_i \delta_{x_i} * B_{hi}.} \quad (\text{ZR})$$

If $J \sim B_h$, write $m_h = \mathbb{E}J$.

Definition 2 (Finite-horizon zero-repair bill).

$$\boxed{\mathcal{B}_n := \inf \left\{ \mathbb{E}_{B_\emptyset} J : \{B_h\}_{|h| \leq n} \text{ satisfies (ZR)} \right\}.}$$

If no feasible depth- n tree exists, the value is $+\infty$.

Operationally, $\mathcal{B}_n$ is the least expected initial nonnegative working inventory which makes every one-step information identity exact without injecting fresh repair after the mismatch has been revealed.

Three quantities must not be conflated:

1. $1 - m_\pi$ is an unmatched block mass in a two-step matching problem;
2. $\|\Psi\|_{\mathbf{X}_m}$ is an auxiliary norm of hidden ordering phase;
3. $\mathcal{B}_n$ is the actual zero-repair inventory bill.

A bounded phase norm is useful only if one also proves a bridge from bounded phase-safe continuation to a bounded increment of $\mathcal{B}_n$.

3 The explicit quartic pair

The principal pair is

$$\boxed{P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64} \right)}$$

on $\{0, 1, 2, 3, 4\}$. It has common mean 2, common variance $3/2$, vanishing odd centered moments, and

$$m_4(P) = \frac{9}{2}, \quad m_4(Q) = \frac{33}{8}.$$

The probability-generating functions satisfy

$$\boxed{G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.}$$

Thus the PGF jets through order 3 agree at $z = 1$ and the first nonzero factorial-moment defect is quartic.

A useful decomposition is

$$G_P(z) = \frac{7}{8}R(z) + \frac{1}{8}E(z), \quad G_Q(z) = \frac{7}{8}R(z) + \frac{1}{8}O(z),$$

where

$$E(z) = \frac{1 + 6z^2 + z^4}{8}, \quad O(z) = \frac{z + z^3}{2},$$

$$R(z) = \frac{7 + 16z + 10z^2 + 16z^3 + 7z^4}{56}.$$

Then

$$E(z) - O(z) = \frac{1}{8}(1 - z)^4.$$

Statically,

$$E(z)^2 - O(z)^2 = \frac{1}{64}(1 - z^2)^4 = \frac{1}{8}(E(z^2) - O(z^2)),$$

but this scalar self-similarity is not a free causal lift.

4 Exact two-step causal obstruction

For a one-step coupling of E and O , define

$$M(d) := \sup_{\gamma \in \Pi(E, O)} \Pr(X - Y = d).$$

Direct diagonal capacity gives

$$M(\pm 1) = \frac{5}{8}, \quad M(\pm 3) = \frac{1}{8}.$$

For a two-step co-adapted coupling, equality of block totals requires

$$X_2 - Y_2 = Y_1 - X_1.$$

Hence the best conditional success probability is at most $5/8$, and the bound is attained.

Theorem 1 (Exact two-step causal block match). *For the parity core,*

$$C_{\text{causal}}^{(2)}(E, O) = \frac{5}{8}.$$

The corresponding static maximal common mass is

$$C_{\text{static}}^{(2)}(E, O) = \frac{7}{8}.$$

Therefore

$$C_{\text{static}}^{(2)} - C_{\text{causal}}^{(2)} = \frac{1}{4}.$$

For one attaining causal coupling, the block discrepancy

$$D = (Y_1 + Y_2) - (X_1 + X_2)$$

has law

$$\Pr(D = 0) = \frac{5}{8}, \quad \Pr(D = \pm 2) = \frac{1}{8}, \quad \Pr(D = \pm 4) = \frac{1}{16}.$$

This is the first explicit sign that scalar common mass is not the full causal resource: ordering information contributes an additional obstruction.

5 First exact residual states

Removing the matched block mass $5/8$, normalizing the residual mass $3/8$, and applying the common lattice scaling $s \mapsto s/2$ gives

$$P_1 = \frac{1}{24}(1, 5, 12, 5, 1), \quad Q_1 = \frac{1}{24}(0, 9, 6, 9, 0).$$

Moreover,

$$G_{P_1}(z) - G_{Q_1}(z) = \frac{1}{24}(1 - z)^4.$$

The next exact two-step causal match is

$$C_{\text{causal}}^{(2)}(P_1, Q_1) = \frac{47}{72}, \quad r_1 = \frac{25}{72}.$$

One exact normalized residual pair is

$$P_2 = \frac{1}{200}(1, 10, 12, 50, 54, 50, 12, 10, 1),$$

$$Q_2 = \frac{1}{200}(0, 0, 44, 28, 56, 28, 44, 0, 0),$$

with

$$G_{P_2}(z) - G_{Q_2}(z) = \frac{(1 - z)^4}{200}(1 + 14z + 18z^2 + 14z^3 + z^4).$$

Thus the pure ray $c(1 - z)^4$ is not invariant, even though the fourth-order zero itself is preserved. For this pair,

$$C_{\text{causal}}^{(2)}(P_2, Q_2) = \frac{14241}{20000}, \quad r_2 = \frac{5759}{20000}.$$

6 Exact reduction to an N^2 transport problem

Let $A = (A_x)$ and $B = (B_y)$ be finite laws on a common integer lattice. For each displacement d , define

$$\alpha_d(A, B) := \max_{\gamma \in \Pi(A, B)} \Pr(X - Y = d).$$

Because the event $X - Y = d$ is one transport diagonal,

$$\alpha_d(A, B) = \sum_t \min\{A_t, B_{t-d}\}.$$

Theorem 2 (Two-step transport reduction).

$$C_{\text{causal}}^{(2)}(A, B) = \max_{\pi \in \Pi(A, B)} \sum_{x, y} \pi_{xy} \alpha_{y-x}(A, B).$$

Put

$$c_d(t) := \min\{A_t, B_{t-d}\}.$$

For a first-stage transport π , the saturated matched block-total measure is

$$M_\pi(s) = \sum_{x, y} \pi_{xy} c_{y-x}(s - x).$$

Let

$$m_\pi := \sum_s M_\pi(s), \quad r := 1 - m_\pi.$$

Before the common affine lattice normalization, the residual laws are

$$\boxed{A_{\text{res}} = \frac{A * A - M_\pi}{r}, \quad B_{\text{res}} = \frac{B * B - M_\pi}{r}.}$$

Distinct feasible transports with the same low-order scalar profile can produce different children. Therefore the causal renormalization is set-valued:

$$\boxed{(A', B') \in \mathcal{R}(A, B).}$$

The implementation applies a *common* affine lattice normalization

$$s \mapsto \frac{s - c}{g}$$

to both residual laws, where c is the first positive common support index and g is the gcd of the positive support offsets. Consequently, the exact child map is piecewise affine in π : it is affine while the support branch (c, g) is fixed, and may kink when the support branch changes.

7 Quartic defect renormalization

Write

$$D(z) = G_A(z) - G_B(z), \quad S(z) = G_A(z) + G_B(z).$$

Before lattice normalization,

$$D_{\text{res}}(z) = \frac{D(z)S(z)}{r}.$$

Therefore a fourth-order zero at $z = 1$ is preserved.

Define the quartic factorial defect

$$\boxed{\eta(A, B) := \sum_i \binom{i}{4} (A_i - B_i) = \frac{1}{4!} (G_A - G_B)^{(4)}(1).}$$

Under the common affine normalization with lattice spacing g ,

$$\boxed{\eta' = \frac{2\eta}{rg^4}, \quad rg^4\eta' = 2\eta.}$$

Let

$$V := \text{Var}(A) = \text{Var}(B), \quad \chi := \frac{\eta}{V^2}.$$

If

$$\tilde{V}' := g^2 V', \quad R := \frac{\tilde{V}'}{V},$$

then

$$\boxed{\frac{\chi'}{\chi} = \frac{2}{rR^2}.}$$

Hence the scale-free critical hypersurface is

$$\boxed{rR^2 = 2.}$$

On it, the quartic mode is marginal.

8 Exact quartic bridge and critical low-order fingerprint

The exact symbolic bridge audit gives

$$\begin{aligned} H(z) &= \frac{z^2(5z^4 + 6z^2 + 5)}{64}, \\ \eta_{\text{br}}(z) &= \frac{5}{64}z^2(1 - z^2)^2, \\ \Phi(z) &= \frac{5z^2(z + 1)^2}{64(z - 1)^2}. \end{aligned}$$

The associated absolute-envelope bookkeeping is

inherited envelope mass = $\frac{5}{14}$,	leftover mass at 0 = $\frac{9}{14}$,	inherited envelope mean = $\frac{5}{3}$.
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The distinguished low-order target is

$T = (m, \mu_2, \mu_4) = \left(\frac{7}{9}, \frac{4}{3}, 6\right).$

It is not an arbitrary fit. The first two coordinates give

$$r = 1 - m = \frac{2}{9}, \quad R = \frac{2 - \mu_2}{r} = 3,$$

so

$rR^2 = 2.$

The fourth moment recursion is

$$M'_4 = \frac{2M_4 + 6 - \mu_4}{rR^2},$$

which on the critical hypersurface becomes

$$M'_4 = M_4 + \frac{6 - \mu_4}{2}.$$

Thus $\mu_4 = 6$ fixes the quartic moment. At the target,

$M'_4 = M_4,$

$M'_6 \leq \frac{1}{3}M_6 + 5M_4,$

$M'_8 \leq \frac{1}{9}M_8 + \frac{28}{9}M_6 + \frac{35}{9}M_4^2.$

The higher-moment leading factors are contracting, whereas the fourth moment is marginal.

Remark 1 (What the target is and is not). *The point $(7/9, 4/3, 6)$ is an exact low-order critical fingerprint of the quartic scale algebra. It is not a proved invariant causal fixed point. The later phase-controlled experiments show that insisting on this same point at every scale is too restrictive.*

9 Rigorous logarithmic lower bound

For an equal-mean finite-support pair define, for $a > 0$,

$$L(a) := \log \frac{G_P(e^{-a})}{G_Q(e^{-a})}, \quad R_L(a) := \log \frac{G_P(e^{-2a})}{G_P(e^{-a})^2}.$$

Whenever $L(a) > 0$, put

$$c(a) := \frac{L(a)}{a[2L(a) + R_L(a)]}.$$

The global Laplace barrier yields

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \mathfrak{c}_L(P, Q) := \sup_{a > 0: L(a) > 0} c(a).$$

For the explicit quartic pair, the elementary exact choice $a = \log 2$ gives

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq 0.00766189365248 \dots$$

and therefore

$$\mathcal{B}_n = \Omega(\log n).$$

A scalar numerical optimization suggests

$$\mathfrak{c}_L(P, Q) \approx 0.01642637054,$$

but that optimized number is not used as a theorem.

10 Exact phase kinematics under block doubling

Let $V_m \cong \mathbb{R}^{2^m}$ be the ordered binary-history space and let Π_m retain only the Hamming layer. Write

$$W_m := \ker \Pi_m, \quad V_m = U_m \oplus W_m,$$

where U_m is a chosen scalar section.

Under block concatenation $V_{2m} \cong V_m \otimes V_m$, let

$$C_m(e_i \otimes e_j) = e_{i+j}.$$

Then

$$\Pi_{2m} = C_m \circ (\Pi_m \otimes \Pi_m).$$

Theorem 3 (Exact phase decomposition under doubling). *For every $m \geq 1$,*

$$W_{2m} = (W_m \otimes W_m) \oplus (W_m \otimes U_m) \oplus (U_m \otimes W_m) \oplus K_m,$$

where

$$\dim K_m = m^2.$$

The sector K_m is genuinely fresh. Even after both half-blocks have been reduced to scalar Hamming layers, the total layer $i + j$ forgets how the weight was split between the two halves. Thus every doubling creates a new anti-diagonal phase sector with m^2 degrees of freedom.

This exact kinematic statement rules out a globally one-dimensional phase ansatz. It does not imply that all m^2 directions are dynamically populated or costly; positivity and causality select a much smaller feasible subset.

11 Exact fixed-block causal phase fiber

Let $\Sigma_m = \Pi_m \otimes \Pi_m$ be the joint scalarization and let $A_m \pi = b_m$ encode the fixed-block causal and marginal constraints. For a scalar profile r , fix any section $\tau_m(r)$ with $\Sigma_m \tau_m(r) = r$.

Theorem 4 (Fixed-block causal phase fiber). *The local causal phase fiber is the bounded polytope*

$$\mathcal{C}_m^{\text{loc}}(r) = \{\theta \in \ker \Sigma_m : A_m \theta = b_m - A_m \tau_m(r), \quad \theta \geq -\tau_m(r)\}.$$

It is nonempty if and only if

$$\langle y, b_m \rangle + \langle z, r \rangle \geq 0$$

for every pair (y, z) satisfying

$$A_m^T y + \Sigma_m^T z \geq 0$$

componentwise. For rational source laws the certificate cone is rational polyhedral.

This closes the *local fixed-block* question. It does not close zero repair. A zero-repair state must land in children that remain feasible at all future depths. The scale-stable descendant subset is the genuinely open object.

12 Reduced fresh phase and the anti-diagonal norm

The full fixed-block history matrices are of size 4^m and cannot be materialized at the audited scales $m = 64, 128, \dots$. The numerical program therefore works with the exact reduced N^2 two-block transport polytope. This reduced model is exact for the saturated two-step renormalization step but is not the full history-level causal phase fiber.

For a split residual matrix $R = (R_{ij})$, define anti-diagonal summation

$$(C_m R)_s := \sum_{i+j=s} R_{ij}.$$

Choose the canonical Euclidean section

$$(S_m v)_{ij} := \frac{v_{i+j}}{\#\{(a, b) : a + b = i + j\}}.$$

Then

$$\Pi_{K_m} R = R - S_m C_m R \in K_m.$$

For the A - and B -side split residuals, define

$$K_A := \Pi_K R_A, \quad K_B := \Pi_K R_B,$$

$$\Psi_+ := \frac{K_A + K_B}{2}, \quad \Psi_- := \frac{K_A - K_B}{2}.$$

Because every K_m element has zero total mass on each anti-diagonal, the natural intrinsic transport seminorm is

$$\|\Psi\|_{\text{aKR}} := \sum_s W_1((\Psi|_{i+j=s})_+, (\Psi|_{i+j=s})_-).$$

Equivalently, on each anti-diagonal this is the one-dimensional cumulative transport formula

$$\|\Psi\|_{\text{aKR}} = \sum_s \sum_k \left| \sum_{i \leq k} \Psi_{i, s-i} \right|.$$

Hence the norm is polyhedral.

A convenient scale-covariant phase norm is

$$\|\Psi\|_{\mathbf{x}_m} := \|\Psi\|_1 + m^2 \|\Psi\|_\infty + \frac{1}{m} \|\Psi\|_{\text{aKR}}.$$

The explicit $m\|\Psi\|_2$ term can be omitted because

$$m\|\Psi\|_2 \leq \sqrt{\|\Psi\|_1 m^2 \|\Psi\|_\infty}.$$

Remark 2 (A correction that matters). *The scalar η -like surcharge packets and the fresh K_m phase are not the same object. The scalar packet channel records a one-dimensional residual signed profile; K_m records the split information killed by anti-diagonal summation. Conflating them would erase the very phase degrees of freedom that control descendant viability.*

13 Fresh-phase scaling: numerical audit

This section is numerical evidence only. The audited baseline starts at the $N = 65$ state where the low-order target is attained essentially exactly.

13.1 Baseline $N = 65$ target and its $N = 129$ child

At $N = 65$,

$$(m, \mu_2, \mu_4) = (0.777777777777506, 1.3333333333337308, 6.000000000033278),$$

with scaled target error

$$3.33 \times 10^{-12}.$$

The associated state has

$$r = 0.222222222222494, \quad R = 2.99999999977913,$$

$$\chi = 0.050000000000254 \mapsto 0.049999999980235.$$

For the concrete baseline child at $N = 129$, the nearest low-order profile is

$$(0.765676457399511, 1.322505221024069, 6.067084933619663),$$

with scaled error

$$0.0121013203783.$$

Thus the exact target is not invariant under this concrete phase choice.

13.2 Fresh-phase metrics

state	$\ \Psi\ _1$	$\ \Psi\ _2$	$\ \Psi\ _\infty$	rank ₉₀	$m^{-1}\ \Psi\ _{\text{aKR}}$
$N65 : \Psi_+$	1.24025	0.033105	0.00367596	4	0.147284
$N65 : \Psi_-$	0.99020	0.033771	0.00386449	2	0.067731
$N129 : \Psi_+$	1.33593	0.019443	0.00106962	4	0.162052
$N129 : \Psi_-$	1.05241	0.020564	0.00110890	2	0.068733

The scale-normalized quantities are

	$N = 65$ ($m = 64$)	$N = 129$ ($m = 128$)
$m\ \Psi_+\ _2$	2.11872	2.48870
$m^2\ \Psi_+\ _\infty$	15.0567	17.5247
$m\ \Psi_-\ _2$	2.16134	2.63221
$m^2\ \Psi_-\ _\infty$	15.8290	18.1682

The observed scaling is consistent with

$$\boxed{\|\Psi\|_1 = O(1), \quad \|\Psi\|_2 = O(m^{-1}), \quad \|\Psi\|_\infty = O(m^{-2}), \quad m^{-1}\|\Psi\|_{\text{aKR}} = O(1).}$$

This is exactly the heuristic expected from m^2 fresh directions carrying amplitude $O(m^{-2})$ each. It is not yet a uniform theorem.

Raw two-dimensional KR grows like the grid diameter and is therefore not the right scale-free quantity. The anti-diagonal normalization removes this artificial geometric growth.

13.3 Scalar common surcharge channel

For the centered common surcharge ζ , the audit gives

	$N = 65$	$N = 129$
common surcharge mass	0.1680771	0.1785042
$\ \zeta\ _1$	0.3231634	0.3481909
$m^{-1}\ \zeta\ _{\text{KR}}$	0.0235798	0.0215226
3-point packet share of L^2 energy	0.6501	0.6569

Thus the scalar channel is also scale-stable in the audited regime, but it is only one projection of the full phase dynamics.

14 Phase is a control variable, not a passive residue

Different parent transports on the same $N = 65$ scalar target fiber produce measurably different $N = 129$ descendant viability. A first random/extremal search changed the scaled child target error from the baseline

$$0.0121013204$$

to

$$\boxed{0.0113790598}$$

without moving the parent away from the scalar target beyond numerical precision. Along a line segment in the target fiber, the error decreased monotonically through

$$0.0121013 \rightarrow 0.0119659 \rightarrow 0.0118324 \rightarrow \cdots \rightarrow 0.0113791.$$

This is direct finite-scale evidence that the hidden phase influences future feasibility.

Further phase-controlled searches reduced the child target deficit to the 10^{-2} range and eventually localized the first obstruction almost entirely in the maximum matched mass. A broader fresh-phase search reached

$$m_*^{\text{child}} \approx 0.76815049, \quad \frac{7}{9} - m_*^{\text{child}} \approx 0.00962729.$$

At that stage, the low-order moment errors were no longer the main bottleneck.

15 Reduced Farkas margin and facial reduction

Work in the scaled profile coordinates

$$y = \left(m, \frac{\mu_2}{2}, \frac{\mu_4}{10}\right), \quad T_s = \left(\frac{7}{9}, \frac{2}{3}, \frac{3}{5}\right).$$

Let $\mathcal{P}(A, B)$ be the reduced first-stage transport polytope and

$$\mathcal{R}_{\text{prof}}(A, B) := F\mathcal{P}(A, B) \subset \mathbb{R}^3$$

its scaled profile image. Define the support function

$$h_{\mathcal{R}}(z) := \max_{\pi \in \mathcal{P}(A, B)} z \cdot F\pi.$$

The reduced profile margin is

$$\delta_{\text{prof}}(y) := \min_{\|z\|_1=1} [h_{\mathcal{R}}(z) - z \cdot y].$$

The marginal dual directions are quotiented out. A naive ambient Farkas margin can be exactly zero simply because the feasible set lies on a proper face of the positive cone. Therefore facial reduction is essential before interpreting the margin as regularity.

Because the profile region is convex,

$$\delta_{\text{prof}}(y) \geq \delta \iff y + \delta[-1, 1]^3 \subset \mathcal{R}_{\text{prof}}(A, B).$$

It is enough to realize the eight cube vertices.

At the exact $N = 65$ target,

$$\delta_{65}^{\text{prof}} = 1.6805127 \times 10^{-3}$$

with worst normalized direction approximately

$$(-0.23546, 0.56028, -0.20426).$$

For the original $N = 129$ nearest-target selector, the margin is essentially zero and the worst direction is

$$z = (1, 0, 0),$$

showing that the selector sits on a pure matched-mass ceiling facet.

16 A constructive dimension-free reduced repair lemma

The reduced profile problem does not require an abstract scale-dependent Hoffman constant once an interior cube is available.

Lemma 1 (Inball repair). *Let C be a convex set of probability couplings and let $F : C \rightarrow \mathbb{R}^3$ be linear. Suppose*

$$y + \delta[-1, 1]^3 \subset F(C)$$

with $\delta > 0$. If $\pi \in C$ has

$$F\pi = y + e, \quad \varepsilon := \|e\|_{\infty},$$

then there exists $\pi' \in C$ with $F\pi' = y$ and

$$\|\pi' - \pi\|_1 \leq \frac{2\varepsilon}{\delta + \varepsilon} \leq \frac{2}{\delta}\varepsilon.$$

Proof. If $e = 0$, take $\pi' = \pi$. Otherwise let

$$y_- = y - \delta \frac{e}{\varepsilon}.$$

By the inball assumption, choose $\pi_- \in C$ with $F\pi_- = y_-$. Set

$$q = \frac{\varepsilon}{\delta + \varepsilon}, \quad \pi' = (1 - q)\pi + q\pi_-.$$

Then $F\pi' = y$. Since two probability measures have L^1 distance at most 2,

$$\|\pi' - \pi\|_1 = q\|\pi_- - \pi\|_1 \leq 2q.$$

□

Thus the reduced metric-regularity constant satisfies the explicit bound

$$H^{\text{red}} \leq \frac{2}{\delta}.$$

This solves the local reduced profile correction problem. It does not solve descendant phase safety or the full history-level zero-repair problem.

17 Three bounded macro-circulation controller modes

A bounded centered rank-one circulation has the form

$$D(i, k) = A_i B_k f(i) g(k),$$

with

$$\mathbb{E}_A f = 0, \quad \mathbb{E}_B g = 0, \quad |f|, |g| \leq 1.$$

Then every row and column sum of D vanishes. If a safe anchor is mixed with a fixed product-coupling reserve, sufficiently small linear combinations of such modes preserve both marginals and positivity.

For three modes $D_{m,1}, D_{m,2}, D_{m,3}$, define the scaled profile Jacobian

$$V_m = \begin{bmatrix} F D_{m,1} & F D_{m,2} & F D_{m,3} \end{bmatrix}.$$

If

$$\sigma_{\min}(V_m) \geq \sigma_0 > 0,$$

then every sufficiently small three-dimensional profile displacement can be repaired by solving

$$a = V_m^{-1} e.$$

Indeed,

$$\|a\|_1 \leq \sqrt{3} \|a\|_2 \leq \frac{\sqrt{3}}{\sigma_0} \|e\|_2 \leq \frac{3}{\sigma_0} \|e\|_\infty.$$

With coefficient budget $\|a\|_1 \leq \alpha$, the guaranteed local cube radius is

$$\delta_{\text{ctrl}} \geq \frac{\alpha \sigma_0}{3}.$$

17.1 Numerically stable bounded controller

The improved default triple is

$$(\text{odd075} \times \text{sign}), \quad (\text{sin1} \times \text{odd075}), \quad (\text{core075} \times \text{cos1}).$$

On the audited q7/margin-aware states,

N	$\sigma_{\min}(V_m)$
65	0.00434106
129	0.0040358146
257	0.0041215354

so the observed floor is

$$0.0040358.$$

This improves the earlier bounded triple by roughly 2.5% on the common three-scale audit.

For the new third mode $\text{core075} \times \text{cos1}$, the normalized phase cost is small and stable:

N	$m^{-1} \ \Delta\Psi_+\ _{\text{aKR}}$	$m^{-1} \ \Delta\Psi_-\ _{\text{aKR}}$
65	0.00634	0.00172
129	0.00593	0.00160
257	0.00577	0.00130

Thus the same modes are simultaneously profile-effective and phase-cheap. The missing theorem is an analytic lower bound

$$\inf_m \sigma_{\min}(V_m) > 0$$

for an explicit bounded mode family.

A separate independently reconstructed baseline orbit with the earlier bounded triple gave

$$0.0045923, \quad 0.0040586, \quad 0.0042179, \quad 0.0040415$$

at $N = 65, 129, 257, 513$, respectively. The exact values differ because the $N = 129$ states in the two audits are produced by different phase policies, but neither audit shows a singular-value collapse.

With $\alpha = 0.2$ and $\sigma_0 \simeq 0.00394$, the crude controller guarantee is

$$\frac{\alpha\sigma_0}{3} \simeq 2.62 \times 10^{-4},$$

which is of the same size as independently observed safe margins.

18 Safe cubes and margin-aware continuation

18.1 An explicit $N = 129$ safe cube

For one $N = 129$ center with reduced inball radius

$$\delta = 4.9 \times 10^{-4},$$

all eight scaled cube corners were realized by explicit feasible transports. The resulting child quantities ranged over

$$r \in [0.231998, 0.232978],$$

$$R \in [2.91521, 2.93597],$$

$$\chi' \in [0.0500048, 0.0505063].$$

Across the same eight corners,

$$m^{-1}\|\Psi_+\|_{\text{aKR}} \approx 0.16237\text{--}0.16324.$$

This is the first concrete reduced *safe cube*: not one lucky trajectory, but an entire three-dimensional local profile neighborhood with feasible lifts.

18.2 Margin-aware $N = 257$ center

A later margin-aware selector found the $N = 257$ center

$$(m, \mu_2, \mu_4) = (0.7719745100, 1.3264509196, 6.0259522932)$$

with scaled target error

$$0.00580327$$

and a certified reduced margin floor approximately

$$2.498 \times 10^{-4}.$$

Its phase metrics were

	Ψ_+	Ψ_-
$\ \Psi\ _1$	1.44186	1.06009
$m\ \Psi\ _2$	3.07562	3.13327
$m^2\ \Psi\ _\infty$	24.7794	26.0873
$m^{-1}\ \Psi\ _{\text{aKR}}$	0.17358	0.06855
rank_{90}	4	2

The resulting next state at $N = 513$ had

$$r = 0.22802549, \quad R = 2.95383240, \quad \chi' = 0.05051886.$$

No explosive phase mode appeared.

At this $N = 513$ stress state, the exact target again lay outside the reachable profile region. The dominant obstruction was again the pure mass facet; no new low-order moment obstruction was exposed in that stress test.

19 Anchor failures that clarified the problem

Several plausible anchors fail for distinct reasons.

Product coupling. It provides a convenient positivity reserve but is not dynamically critical. Its iterated χ decays rapidly,

$$0.125 \rightarrow 0.0579 \rightarrow 0.0273 \rightarrow \cdots \rightarrow 8.3 \times 10^{-4},$$

so the product coupling leaves the quartic critical regime.

Maximum-entropy coupling. At $N = 65$, the maximum-entropy coupling can realize the exact low-order target with a smooth phase and gives essentially

$$r = \frac{2}{9}, \quad R = 3, \quad \chi' = 0.05.$$

But its child target deficit is about 0.01319, so entropy maximization alone does not select the dynamically correct phase.

Parent interiority alone. A parent coupling can sit exactly at the scalar target with a positive local profile margin of order 10^{-3} while its child has fixed-target margin about

$$-0.01525.$$

Thus

$$\boxed{\text{parent interior margin} \not\Rightarrow \text{descendant interior margin.}}$$

This negative control is crucial. A safe anchor must be defined by descendant viability, not by local parent regularity alone.

20 Reproducibility corrections and independent audit

A standalone reimplementer originally standardized block totals by

$$\sqrt{\text{Var } A + \text{Var } B}.$$

The project profile map actually uses

$$\boxed{V = \frac{\text{Var } A + \text{Var } B}{2}, \quad \sigma = \sqrt{V}, \quad z_{\text{block}} = \frac{s - 2\mu}{\sigma}.}$$

The incorrect denominator produced an exact factor 2 error in the second moment and factor 4 in the fourth moment at the symmetric $N = 65$ state. Once corrected, the independently reconstructed $N = 65$ feature matrix and controller singular values agreed with the project implementation to roughly 10^{-13} .

The same independent audit reconstructed the $N = 65$ bootstrap state exactly and then regenerated the archived $N = 129$ child with zero difference in the stored A, B arrays. This is a strong reproducibility check on the actual child map rather than a toy surrogate.

A second implementation issue was exposed at $N = 129$: HiGHS presolve with loose default tolerances can falsely report infeasibility on a badly scaled transport polytope that is demonstrably feasible. The production renormalization solver therefore prioritizes no-presolve delta-coordinate LPs with tighter primal/dual feasibility tolerances and only then uses fallback configurations.

The independent phase-fiber exploration also reproduced the qualitative fact that different exact-target parent transports have different child margins. Its first random/extremal sample found margins ranging roughly from -0.0111 to -0.0185 around the fixed target. This was not the best known phase selector, but it independently confirms that descendant viability is phase-sensitive.

21 Fixed-target optimization and the Farkas ridge

The exact low-order target is a useful diagnostic but eventually becomes too restrictive. Before abandoning it, the target fiber was attacked directly.

21.1 Dual-guided child mass maximization

Starting from a q7 exact-target parent, the child maximum matched mass was improved to

$$m_*^{\text{child}} = 0.7703034826,$$

so the pure mass deficit became

$$\frac{7}{9} - m_*^{\text{child}} = 0.0074742951.$$

This was a substantial improvement over the earlier 0.0096273 deficit.

However, the full Farkas audit then exposed a new active facet. The worst normalized direction was approximately

$$z = (0.90449, 0, -0.09551),$$

and the exact fixed-target margin was

$$-0.00811947.$$

Thus once the pure mass ceiling was pushed back, a mixed mass–quartic facet became dominant.

21.2 Farkas-guided phase step

Optimizing directly against the active Farkas support plane improved the exact fixed-target margin to

$$-0.00732941.$$

The worst direction moved to

$$z \approx (0.92273, 0, -0.07727),$$

while the pure mass facet was almost tied at approximately

$$-0.00725070.$$

The two active constraints were therefore nearly balanced.

A max–min step using both active-facet subgradients found a common local improvement in the linearization, but the exact objective crossed a nearby piecewise-linear kink. A micro-line search improved the margin only at the 10^{-8} level. This is strong evidence that the optimized point lies near a nonsmooth ridge of the *fixed-target* objective. It is not a global certificate of fixed-target impossibility.

22 The decisive correction: invariant tube, not fixed point

The negative fixed-target margin does not refute the desired upper-bound mechanism. It refutes only the unnecessarily strong demand that the same three-dimensional target point remain feasible at every scale.

Define the scalar target deficit and descendant safety slack as different quantities:

$$d_m(\pi) := \text{dist}_{\text{prof}}(F(\text{child}(\pi)), T_s),$$

$$\rho_m(\pi) := \text{dist}(\text{child}(\pi), \partial \mathfrak{K}_{2m}).$$

For a target tolerance ε and phase budget C_Ψ , define

$$\rho_m^*(\varepsilon, C_\Psi) := \sup \{ \rho_m(\pi) : d_m(\pi) \leq \varepsilon, \|\Psi(\pi)\|_{\mathbf{x}_m} \leq C_\Psi \}.$$

The correct uniform numerical/theorem target is

$$\inf_{m=2^k} \rho_m^*(\varepsilon_0, C_\Psi) \geq \rho_0 > 0.$$

22.1 Positive child tube after fixed-target failure

At the best Farkas-guided child, the exact target itself is outside the reachable profile region, but nearby interior centers exist. Solving

$$\min_y \|y - T_s\|_\infty \quad \text{subject to} \quad y + \delta[-1, 1]^3 \subset \mathcal{R}_{\text{prof}}^{\text{child}}$$

gives the following finite-LP certificates:

required inball δ	smallest scaled target error
2.5×10^{-4}	0.0076493
5.0×10^{-4}	0.0080133
1.0×10^{-3}	0.0089447
1.5×10^{-3}	0.0114313

In particular, with the provisional critical-band tolerance

$$\varepsilon_0 = 10^{-2},$$

one can retain an interior reduced child cube of at least

$$\rho_0 \approx 10^{-3}$$

at this transition. The precise statement is a finite numerical certificate inside the reduced N^2 model, not an asymptotic theorem.

This is the central change of viewpoint:

$$T_s \notin \mathcal{R}_{\text{child}} \quad \text{can coexist with} \quad \exists y_{\text{safe}} \approx T_s : y_{\text{safe}} + \rho_0[-1, 1]^3 \subset \mathcal{R}_{\text{child}}.$$

The fixed point can fail while the invariant tube remains viable.

23 Candidate phase-augmented invariant tube

A useful working form is

$$\begin{aligned} \mathfrak{K}_m = \{ \Theta : & \|y(\Theta) - T_s\|_\infty \leq \varepsilon_0, \\ & \delta_{\text{prof}}(y(\Theta)) \geq \rho_0, \\ & \chi(\Theta) \in I_\chi, \quad rR^2 \in I_K, \\ & \widetilde{M}_6 \leq C_6, \quad \widetilde{M}_8 \leq C_8, \\ & \|\Psi_+\|_{\mathbf{x}_m} \leq C_+, \\ & \|\Psi_-\|_{\mathbf{x}_m} \leq C_-, \\ & m^{-1}\|\zeta\|_{\text{KR}} \leq C_0, \\ & \text{positivity/support branch regularity holds} \}. \end{aligned}$$

The current numerical working scales are

$$\boxed{\varepsilon_0 \sim 10^{-2}, \quad \rho_0 \sim 10^{-3}.}$$

These constants are not theorem constants. They summarize the audited finite-scale geometry and are intended as targets for the next invariant-tube iteration.

24 The theorem that would close the upper bound

Conjecture 1 (Uniform Invariant-Tube / Safe-Anchor Lemma). *There exist constants*

$$\varepsilon_0, \rho_0, \sigma_0, C_\Psi, C_D, C_{\text{bill}} > 0$$

independent of the dyadic scale m , and phase-augmented invariant sets $\mathfrak{K}_m$, such that every state $\Theta_m \in \mathfrak{K}_m$ admits a prefix-compatible causal lift π_m^ with child*

$$\Theta_{2m} = \mathcal{R}_m(\Theta_m, \pi_m^*) \in \mathfrak{K}_{2m},$$

with uniform descendant slack

$$\boxed{\text{dist}(\Theta_{2m}, \partial\mathfrak{K}_{2m}) \geq \rho_0,}$$

and uniformly bounded additional zero-repair inventory

$$\boxed{\Delta\mathcal{B}_m \leq C_{\text{bill}}.}$$

Moreover, each state admits three bounded centered circulation modes $D_{m,1}, D_{m,2}, D_{m,3}$ satisfying

$$\boxed{\sigma_{\min} \begin{bmatrix} F_m D_{m,1} & F_m D_{m,2} & F_m D_{m,3} \end{bmatrix} \geq \sigma_0}$$

and

$$\boxed{\|\widehat{\Psi}(D_{m,j})\|_{\mathbf{x}_m} \leq C_D.}$$

If this lemma is proved, the local controller and inball-repair machinery provide a uniform finite-dimensional correction step, while the inventory bound yields

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + C_{\text{bill}}.$$

Dyadic iteration then gives

$$\mathcal{B}_n = O(\log n),$$

and the rigorous lower bound supplies

$$\boxed{\mathcal{B}_n = \Theta(\log n).}$$

The load-bearing distinction is that the phase norm itself is not the bill. One still has to prove the bridge

$$\boxed{\text{bounded safe phase/state transition} \implies \Delta\mathcal{B}_m \leq C_{\text{bill}}.}$$

Without this implication, bounded $\mathbf{X}_m$ is only an auxiliary compactness statement.

25 Relation to existing theory

Two neighboring theories are directly relevant but do not by themselves solve the present problem.

25.1 Causal transport

Discrete-time causal transport already has a dynamic-programming theory. In particular, Backhoff, Beiglbock, Lin and Zalashko establish a dynamic programming principle for causal transport plans. This supports the general view that prefix restrictions, compactness and measurable/recursive selection should be treated with existing adapted-transport machinery rather than reinvented from scratch. The missing part here is more specialized: a scale-uniform positive phase continuation theorem for the present zero-repair contract.

25.2 KMT strong approximation

The Komlos–Major–Tusnady strong approximation and later proofs, including Chatterjee’s Stein-inspired strong-embedding argument, show that logarithmic multiscale discrepancy control is natural in strong coupling problems. The analogy is useful at the architectural level: dyadic construction plus a scale-independent correction at each doubling is exactly the pattern that would produce $O(\log n)$ here.

However, “strong embedding” in KMT is not the same as the present nonanticipating zero-repair identity. Standard KMT gives a joint coupling with a small global sup discrepancy; it does not automatically provide the prefix-compatible buffer law

$$\mu_Q * B_h = \sum_i p_i \delta_{x_i} * B_{hi}$$

with no post-mismatch repair. Therefore KMT is a source of proof architecture, not a theorem that can currently be imported as the desired upper bound.

26 What has been proved, what is numerical, what is open

A. Rigorous / exact statements

1. The explicit source pair has PGF defect $(1 - z)^4/64$ and matching jets through order 3.
2. The parity core satisfies $E - O = \frac{1}{8}(1 - z)^4$.
3. $C_{\text{causal}}^{(2)}(E, O) = 5/8$ while the static common mass is $7/8$.
4. The first residual states P_1, Q_1 and P_2, Q_2 above are exact.
5. The saturated two-step co-adapted problem reduces exactly to an N^2 transport LP.
6. The residual difference has the exact multiplicative renormalization $D_{\text{res}} = DS/r$ before common lattice normalization.
7. The quartic intensity satisfies $rg^4\eta' = 2\eta$.
8. The scale-free quartic coordinate satisfies $\chi'/\chi = 2/(rR^2)$.
9. The critical low-order algebra gives $(m, \mu_2, \mu_4) = (7/9, 4/3, 6)$, $r = 2/9$, $R = 3$, and $rR^2 = 2$.
10. At this target, M_4 is marginal and the displayed M_6, M_8 upper recursions have contracting leading factors.
11. The exact phase decomposition under doubling contains a fresh sector K_m with $\dim K_m = m^2$.
12. Every fixed-block local causal phase fiber has an exact finite polyhedral/Farkas characterization.
13. The reduced anti-diagonal KR norm is polyhedral.
14. An interior reduced profile cube gives the explicit dimension-free repair estimate $H^{\text{red}} \leq 2/\delta$.
15. The Laplace barrier proves $\mathcal{B}_n = \Omega(\log n)$. The explicit lower coefficient from $a = \log 2$ is 0.00766189365248....

B. Numerical / finite-LP evidence

1. The exact low-order target is attained at the audited $N = 65$ state but not at its concrete $N = 129$ descendant.
2. Fresh phase obeys the observed scale-covariant pattern $L^1 = O(1)$, $L^2 = O(m^{-1})$, $L^\infty = O(m^{-2})$ and $m^{-1}\text{aKR} = O(1)$ on the audited scales.
3. The scalar common surcharge channel is also stable after scale normalization.
4. Different parent phases on the same scalar target fiber produce different child viability.
5. Reduced Farkas geometry first exposes a pure mass ceiling and then, after mass optimization, a mixed mass-quartic active facet.
6. The improved bounded three-mode controller remains nonsingular on the audited scales and has bounded normalized phase cost.
7. Explicit safe cubes exist in the reduced N^2 model.
8. Margin-aware continuation reaches $N = 257$ and a critical-looking $N = 513$ state without a phase blow-up.
9. Product and maximum-entropy anchors are individually inadequate dynamic policies.
10. Parent local interiority does not imply descendant interiority.
11. Dual/Farkas-guided optimization appears close to a nonsmooth ridge on the exact-target parent fiber.
12. Most importantly, after the exact target fails, nearby centers with positive inball margin remain available: the invariant tube survives.

C. Open theorem-level statements

1. Prove an analytic scale-uniform lower bound $\inf_m \sigma_{\min}(V_m) > 0$ for an explicit bounded controller triple.
2. Prove uniform bounds on the full phase norm and scalar surcharge over an entire invariant tube, not only along selected trajectories.
3. Control support-branch changes (c, g) and positivity uniformly.
4. Upgrade the reduced N^2 safe-cube picture to the full history-level, prefix-compatible zero-repair phase state.
5. Prove existence of a descendant-safe anchor with $\rho_m^*(\varepsilon_0, C_\Psi) \geq \rho_0 > 0$ for every dyadic scale.
6. Prove the missing inventory bridge $\Delta\mathcal{B}_m \leq C_{\text{bill}}$.
7. Deduce the bounded doubling estimate $\mathcal{B}_{2n} \leq \mathcal{B}_n + C$.
8. Deduce the matching upper bound $\mathcal{B}_n = O(\log n)$ and hence $\mathcal{B}_n = \Theta(\log n)$.
9. Determine the exact asymptotic constant of $\mathcal{B}_n / \log n$.

27 Recommended next proof attack

The next step should not be another attempt to force the exact target point. The appropriate finite-scale experiment and analytic target are now the same: construct an *invariant-tube orbit*.

Fix provisional constants, for example

$$\varepsilon_0 = 10^{-2}, \quad \rho_0 = 10^{-3},$$

and at each audited scale solve for a parent phase-safe coupling such that the child admits a center satisfying

$$\|y_{2m} - T_s\|_\infty \leq \varepsilon_0, \quad \delta_{\text{prof}}(y_{2m}) \geq \rho_0,$$

while

$$\|\Psi_{2m,\pm}\|_{\mathbf{x}_{2m}} \leq C_\pm.$$

The numerical objective should therefore maximize descendant slack, not minimize target deficit:

$$\boxed{\max_{\pi} \rho_m(\pi) \quad \text{subject to} \quad d_m(\pi) \leq \varepsilon_0, \quad \|\Psi(\pi)\|_{\mathbf{x}_m} \leq C_{\Psi}.}$$

On a fixed common-affine support branch, the parent residual laws are affine in π . This opens the possibility of a single joint LP containing the parent transport and the eight child safe-cube witness transports. If the support branch remains stable, this is preferable to a generic bilevel search. If the support branch changes, the problem can be split into finitely many affine branches.

The analytic route should then identify the limiting three-mode controller Jacobian and prove its determinant is bounded away from zero, while deriving a uniform phase/inventory estimate for the selected safe anchors. This is the most direct path from the present finite certificates to the desired bounded doubling theorem.

28 Reproducibility ledger

The current computational chain is intentionally modular. The following files form the principal audit trail.

File	Role
<code>wpb_causal_renorm_targetchi_audit_fixed_v3.py</code>	Core reduced transport, matched measure, residual child map, common affine normalization, renormalization identity audits.
<code>wpb_causal_moment_region_attack.py</code>	Exact reduced profile map (m, μ_2, μ_4) in standardized one-step coordinates.
<code>wpb_target_moments_fixedpoint_orbit.py</code>	Low-order target-orbit crash test.
<code>wpb_quartic_bridge_attack_exact.py</code>	Symbolic inherited bridge, critical target, exact moment recursion checks.
<code>wpb_fresh_phase_audit.py</code>	Canonical fresh K_m projection and phase metrics.
<code>wpb_fresh_phase_audit_results.json</code>	$N = 65/129$ baseline phase and surcharge audit.
<code>wpb_fresh_phase_audit_states.npz</code>	Reproducible baseline $N = 65/129$ states.
<code>wpb_phase_controlled_bilevel.py</code>	Early phase-fiber descendant-viability search.
<code>wpb_phase_q7_refine.py</code> / <code>wpb_phase_q7_refine_best.npz</code>	Improved fresh-phase selector and checkpoint.
<code>wpb_profile_farkas_margin_audit.py</code>	Exact reduced profile support/Farkas margin oracle.
<code>wpb_reduced_farkas_margin_audit.py</code>	Facial-reduction diagnostics; explicitly scoped to the reduced N^2 problem.
<code>wpb_margin_constrained_selector.py</code>	Margin-aware interior-center continuation.
<code>wpb_safe_cube_audit.py</code>	Eight-corner reduced safe-cube verification.
<code>wpb_macro_circulation_audit.py</code>	Search for bounded rank-one controller triples.
<code>wpb_macro_controller_phase_corecos.py</code>	Phase-cost audit for the improved <code>core075×cos1</code> triple.
<code>wpb_margin_N257_active_center_state.json/.npz</code>	Margin-aware $N = 257$ anchor and child state.
<code>wpb_dual_guided_phase.py</code>	Child mass dual-guided phase optimization.
<code>wpb_farkas_guided_phase.py</code>	Direct active-Farkas-facet phase optimization.

File	Role
<code>wpb_farkas_maxmin_step.py</code>	Nonsmooth two-active-facet max–min step.
<code>wpb_farkas_microstep.py</code>	Micro-line-search across the local Farkas kink.
<code>wpb_farkas_child_tube_results.json</code>	Child invariant-tube center/inball certificates.
<code>wpb_maxent_anchor.py</code>	Maximum-entropy anchor test.
<code>wpb_product_anchor_orbit.py</code>	Product-coupling dynamic crash test.
<code>wpb_causal_phase_replicable_audit_fixed.py</code>	Corrected standalone profile/controller verifier.
<code>replicable_fixed_N65.json</code>	Standalone $N = 65$ local margin and descendant crash test.
<code>replicable_fixed_controller_4scales.json</code>	Independently reconstructed controller singular values through $N = 513$.
<code>state_N65.json, state_N129.json</code>	Portable probability-vector states for external verification.

29 Final checkpoint statement

The quartic zero-repair program has passed the point where the main ambiguity is “what is the missing variable?” The missing variable is phase, its fresh anti-diagonal sector is explicit, and its finite-scale influence on descendant viability is directly measurable.

It has also passed the point where the main open problem is merely local feasibility. Local fixed-block feasibility is polyhedral, the reduced profile region has an exact Farkas oracle, and an interior cube gives an explicit scale-free local repair estimate.

The remaining obstruction is genuinely dynamical:

select a prefix-compatible, phase-safe anchor at every dyadic scale,

keep the full state inside a uniform critical tube,

and pay only $O(1)$ additional inventory per doubling.

The exact target fixed point is not the correct invariant. The current finite certificates instead support a robust invariant-tube picture. If the Uniform Invariant-Tube / Safe-Anchor Lemma is proved with a bounded inventory increment, then the remaining argument is immediate:

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + C \implies \mathcal{B}_n = O(\log n)$$

and therefore, using the already proved lower bound,

$$\mathcal{B}_n = \Theta(\log n).$$

At the date of this checkpoint, that final implication is conditional. The project is not QED.

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